1 I

1.1 (1)

Generally, the equations of motion for A and B are, respectively:

$$m_A\ddot{x}_A = -k_A x_A + k_{AB}(x_B - x_A),$$
$$m_B\ddot{x}_B = -k_B x_B - k_{AB}(x_B - x_A).$$

Under the conditions $m_A = m_B = m$ and $k_A = k_B = k$, these equations become:

$$m\ddot{x}_A = -kx_A + k_{AB}(x_B - x_A),$$
$$m\ddot{x}_B = -kx_B - k_{AB}(x_B - x_A).$$

1.2 (2)

Let $x_{A+B} = x_A + x_B$ and $x_{A-B} = x_A - x_B$. By adding and subtracting the equations of motion, respectively, we obtain:

$$\ddot{x}_{A+B} = -\frac{k}{m}x_{A+B},$$
$$\ddot{x}_{A-B} = -\frac{k + 2k_{AB}}{m}x_{A-B}.$$

Thus, the corresponding angular frequencies are:

$$\omega_{A+B} = \sqrt{\frac{k}{m}},$$
$$\omega_{A-B} = \sqrt{\frac{k + 2k_{AB}}{m}}.$$

2 II

2.1 (3)

When A and B are not connected, they behave as independent harmonic oscillators. Since they oscillate with the same angular frequency ω, we have:

$$\frac{k_A}{m_A} = \frac{k_B}{m_B} = \omega^2.$$

2.2 (4)

The Lagrangian of this system is:

$$L = \frac{1}{2}m_A\dot{x}_A^2 + \frac{1}{2}m_B\dot{x}_B^2 - \frac{1}{2}k_A x_A^2 - \frac{1}{2}k_B x_B^2 - \frac{1}{2}k_{AB}(x_A - x_B)^2.$$

2.3 (5)

From the definitions of the center-of-mass coordinate and the relative coordinate, we have:

$$x_A = x_G + \frac{m_B}{m_G} x_r,$$
$$x_B = x_G - \frac{m_A}{m_G} x_r.$$

Therefore, the kinetic and potential energy terms can be rewritten as:

$$\frac{1}{2} m_A \dot{x}_A^2 + \frac{1}{2} m_B \dot{x}_B^2 = \frac{1}{2} m_G \dot{x}_G^2 + \frac{1}{2} m_r \dot{x}_r^2,$$
$$\frac{1}{2} m_A x_A^2 + \frac{1}{2} m_B x_B^2 = \frac{1}{2} m_G x_G^2 + \frac{1}{2} m_r x_r^2.$$

Since $k_A/m_A = k_B/m_B = \omega^2$, we can finally rewrite the Lagrangian as:

$$L = \frac{1}{2} m_G \dot{x}_G^2 + \frac{1}{2} m_r \dot{x}_r^2 - \frac{1}{2} m_G \omega^2 x_G^2 - \frac{1}{2} m_r \omega^2 x_r^2 - \frac{1}{2} k_{AB} x_r^2.$$

2.4 (6)

Applying the Euler–Lagrange equation to x_r, we obtain:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}_r}\right) - \frac{\partial L}{\partial x_r} = 0.$$

Substituting the Lagrangian obtained in (5), we get:

$$m_r \ddot{x}_r + m_r \omega^2 x_r + k_{AB} x_r = 0.$$

Therefore, the equation of relative motion is:

$$\ddot{x}_r + \left(\omega^2 + \frac{k_{AB}}{m_r}\right) x_r = 0.$$

2.5 (7)

The angular frequency of the relative motion is:

$$\omega_r = \sqrt{\omega^2 + \frac{k_{AB}}{m_r}}.$$

2.6 (8)

Similarly, applying the Euler–Lagrange equation to x_G, we obtain:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}_G}\right) - \frac{\partial L}{\partial x_G} = 0.$$

Thus:

$$m_G \ddot{x}_G + m_G \omega^2 x_G = 0.$$

According to the initial conditions $x_A(0) = x_0$, $x_B(0) = 0$, and $\dot{x}_A(0) = \dot{x}_B(0) = 0$, we have:

$$\begin{aligned} x_r(0) &= x_0, \\ x_G(0) &= \frac{m_A}{m_A + m_B} x_0, \\ \dot{x}_r(0) &= \dot{x}_G(0) = 0. \end{aligned}$$

Solving the equations of motion under these initial conditions gives:

$$\begin{aligned} x_r(t) &= x_0 \cos(\omega_r t), \\ x_G(t) &= \frac{m_A}{m_A + m_B} x_0 \cos(\omega t). \end{aligned}$$

From the definitions of x_G and x_r, we can express x_A and x_B as:

$$\begin{aligned} x_A &= x_G + \frac{m_B}{m_A + m_B} x_r, \\ x_B &= x_G - \frac{m_A}{m_A + m_B} x_r. \end{aligned}$$

Thus, we obtain:

$$\begin{aligned} x_A(t) &= \frac{m_A x_0}{m_A + m_B} \cos(\omega t) + \frac{m_B x_0}{m_A + m_B} \cos(\omega_r t), \\ x_B(t) &= \frac{m_A x_0}{m_A + m_B} \cos(\omega t) - \frac{m_A x_0}{m_A + m_B} \cos(\omega_r t). \end{aligned}$$

2.7 (9)

When $m_A \gg m_B$, we have:

$$m_G \approx m_A, \qquad m_r \approx m_B.$$

Since $k_B/m_B = \omega^2$, the angular frequency of the relative motion can be approximated as:

$$\begin{aligned} \omega_r &= \sqrt{\omega^2 + \frac{k_{AB}}{m_r}} \\ &\approx \sqrt{\omega^2 + \frac{k_{AB}}{m_B}} \\ &= \omega \sqrt{1 + \frac{k_{AB}}{k_B}} \\ &\approx \omega \left(1 + \frac{k_{AB}}{2k_B}\right). \end{aligned}$$

Here, we have used the condition $k_{AB}/k_B \ll 1$.

Substituting $m_A \gg m_B$ into the results of (8), we obtain:

$$\begin{aligned} x_A(t) &\approx x_0 \cos(\omega t), \\ x_B(t) &\approx x_0 \cos(\omega t) - x_0 \cos(\omega_r t) \\ &= -2x_0 \sin\left(\frac{\omega + \omega_r}{2} t\right) \sin\left(\frac{\omega - \omega_r}{2} t\right). \end{aligned}$$

Therefore, A approximately performs simple harmonic motion with a constant amplitude x_0 and angular frequency ω.

On the other hand, B exhibits a beat-like motion. Its rapid oscillation has an angular frequency approximately equal to $(\omega + \omega_r)/2$, while its amplitude is slowly modulated by the frequency difference:

$$A_B(t) = 2|x_0| \left| \sin\left(\frac{\omega - \omega_r}{2} t\right) \right|.$$

The sign change of the modulation factor corresponds to a phase shift of π in the rapid oscillation.